THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · MEDIUM LEVEL UNIT TEST SET 2

Chapters 8–11: **Introduction to Trigonometry, Some Applications of Trigonometry, Circles, Areas Related to Circles** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. Two concentric circles have radii 14 cm and 7 cm. A sector of angle 60° is marked in both. The area of the region between the two arcs within the sector is: 1

(a) 38.5 cm² **(b)** 154 cm² **(c)** 77 cm² **(d)** 462 cm²

2. If $\tan\theta = \frac{4}{3}$, then $\frac{\sin\theta + \cos\theta}{\cos\theta} =$ 1

(a) \frac{3}{7} **(b)** \frac{7}{3} **(c)** \frac{4}{3} **(d)** \frac{1}{3}

3. A storm breaks a tree at a height of 3 m above the ground. The broken part bends over, its top touches the ground and the broken part makes an angle of 60° with the ground. The original height of the tree is: 1

(a) 6 m **(b)** $3 + 2\sqrt{3}$ m **(c)** 5 m **(d)** $2\sqrt{3}$ m

4. In $\triangle ABC$, right-angled at B, $AB = 11$ cm and $BC = 60$ cm. A circle is inscribed in the triangle. The radius of the inscribed circle is: 1

(a) 30 cm **(b)** 6 cm **(c)** 10 cm **(d)** 5 cm

5. An arc of length 22 cm subtends an angle at the centre of a circle of radius 14 cm. The angle subtended by the arc at the centre is: 1

(a) 180° **(b)** 45° **(c)** 90° **(d)** 91°

6. If $\sin 20^\circ = \cos 70^\circ$ (acute angles), then $a + b =$ 1

(a) 90° **(b)** 60° **(c)** 180° **(d)** 45°

7. The string of a kite is 10 m long. It is taut and makes an angle of 60° with the level ground. The height of the kite above the ground is: 1

(a) 5 m **(b)** $5\sqrt{3}$ m **(c)** $10\sqrt{3}$ m **(d)** $\frac{5\sqrt{3}}{2}$ m

8. Tangents QA and QB are drawn from an external point Q to a circle, and $QA = (8x + 3)$ cm and $QB = (5x + 18)$ cm. The length of QA is: 1

(a) 58 cm **(b)** 5 cm **(c)** 43 cm **(d)** 45 cm

9. A chord of a circle of radius 49 cm subtends a right angle at the centre. The area of the minor segment is: 1

(a) 1886.5 cm² **(b)** 1286.25 cm² **(c)** 686 cm² **(d)** 1200.5 cm²

10. The value of $2\sin^2 60^\circ - \tan^2 30^\circ$ is: 1

(a) $\frac{19}{6}$ **(b)** $\frac{13}{6}$ **(c)** $\frac{7}{6}$ **(d)** $\frac{1}{6}$

11. A statue 14 m tall stands on a pedestal. From a point on the ground, the angle of elevation of the top of the statue is 60° and that of the top of the pedestal is 45°. The height of the pedestal is: 1

(a) $\frac{14\sqrt{3}}{3}$ m **(b)** 14 m **(c)** $7+7\sqrt{3}$ m **(d)** $14\sqrt{3}$ m

12. PA and PB are tangents from an external point P to a circle, and $\angle PAB = 35^\circ$. The $\angle APB$ is: 1

(a) 70° **(b)** 55° **(c)** 35° **(d)** 110°

13. A wire in the form of a circle of radius 21 cm is bent into a square. The area of the square is: 1

(a) 1386 cm² **(b)** 4356 cm² **(c)** 33 cm² **(d)** 1089 cm²

14. If $\sin\theta = \frac{12}{13}$, then $\sec\theta + \tan\theta =$ 1

(a) \frac{1}{5} **(b)** \frac{25}{12} **(c)** 5 **(d)** \frac{13}{5}

15. If the angle of elevation of the top of a tower is 30° at a point P and it is 60° at a point nearer to the tower, then the point nearer to the tower is at: 1

(a) half the distance of P from the foot

(b) twice the distance of P from the foot

(c) the same distance as P

(d) one-third of the distance of P from the foot

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. A boy whose eyes are 2 m above the ground stands 24 m away from a vertical building. The angle of elevation of the top of the building from his eyes is 60°. Find the height of the building. 2

17. PA and PB are tangents from an external point P to a circle with centre O, touching it at A and B. $\angle APB = 36^\circ$. Find the $\angle OAB$. 2

18. A horse is tied at the midpoint of one side of a large square field with a rope 14 m long. Find the area of the field it can graze. 2

19. Evaluate: $\sin^2 45^\circ + \cos^2 45^\circ$. 2

20. From the top of a tower 9 m high, the angles of depression of two objects on level ground, on the same side of the tower and in line with its foot, are 45° and 60°. Find the distance between the two objects. 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. The radii of two circles are 14 cm and 7 cm. Find the radius of the circle whose circumference equals the sum of their circumferences. 3

22. Prove that: $\sqrt{1-\sin^2 A}\cdot\sec A = 1$. 3

23. A flagstaff stands on the top of a building 6 m high. From a point on level ground the angles of elevation of the bottom and the top of the flagstaff are 45° and 60° respectively. Find the length of the flagstaff. 3

24. Tangents PA and PB are drawn from an external point P to a circle of centre O and radius 30 cm. The length of each tangent is 16 cm. Find the area of the quadrilateral $PAOB$ (in cm^2). 3

OR

The radii of two circles are 3 cm and 4 cm. Find the radius of the circle whose area equals the sum of their areas. 3

25. If $\tan(2x^\circ) = \sqrt{3}$, find the value of x (an acute angle). 3

OR

The angle of elevation of the top of a vertical tower from a point on level ground is 45°. After walking 8 m towards the foot of the tower along a straight line, the angle of elevation becomes 60°. Find the height of the tower. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. If $\tan A = \dfrac{15}{8}$, evaluate $\dfrac{3\sin A + 2\cos A}{4\sin A + 1\cos A}$. 5

OR

The angle of elevation of the top of a vertical tower from a point on level ground is 30°. After walking 2 m towards the foot of the tower along a straight line, the angle of elevation becomes 45°. Find the height of the tower and the initial distance of the point from its foot (in this order). 5

27. Prove that if PA and PB are tangents from P to a circle with centre O, then $\angle APB + \angle AOB = 180^\circ$. 5

OR

An umbrella has 8 ribs which are equally spaced and the umbrella, when opened flat, is a circle of radius 42 cm. Find the area between two consecutive ribs. 5

END OF QUESTION PAPER — **Total: 50 Marks**

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